

① ④ evolve $ev(0, \tau) / ev(\tau, T)$



dressing OFF

interaction zone

$R_{CZ} = 2.5 \mu m$

gate ON

gate zone

Timing diagram of a quantum control sequence. The vertical axis shows time in units of τ , with markers at 0, 2.085 (τ), 54.47, 55.16, 107.5, and 110.5 (τ). The sequence is divided into four numbered steps: 1 (blue), 2 (orange), 3 (red), and 4 (blue).

Step 1 (blue): dressing, drive I, drive Q (200 ns excerpt).

Step 2 (orange): move-AOD x, move-AOD y, tone $f(t)$.

Step 3 (red): gate, frame a,b (R_z ticks).

Step 4 (blue): dressing, drive I, drive Q (200 ns excerpt), meas.

Legend for step activation:

	1	2	3	4
move-AOD x	✓	✓		
move-AOD y		✓		
addr-AOD x	✓			✓
addr-AOD y				✓
drive I	✓			✓
drive Q	✓			✓
dressing	✓			✓
gate			✓	

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seg    semantic    wall clock (μs)
      terms (provenance) · frame · transport · ins

seg0   [0,τ)      [ 0.000,  2.085]
      Ω·Xa Ω·Xb 0J·ZaZb (src) · - · - · -

seg1   -          [ 2.085, 54.466]
      - · - · a,b-gate · -

seg2   [0,π/4)    [ 54.466, 55.162]
      s·ZaZb (insertion) · Rz(s·0.25π) a,b · - · - INS

seg3   -          [ 55.162, 107.543]
      - · - · a,b-int · -

seg4   [τ,T)      [107.543, 110.458]
      Ω·Xa Ω·Xb 0J·ZaZb (src) · - · - · -

branch total 110.5 μs = 5.0 evolve + 104.8 transport + 0.70 gate
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